SHAKING-TABLE TESTS OF A BART RAILCAR

Earthquake early warning, resilience and shaking-table experiments

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PEER Center · UC Berkeley



Laboratory demonstrations | November 9–10, 2026

Monday, November 9 | Agenda

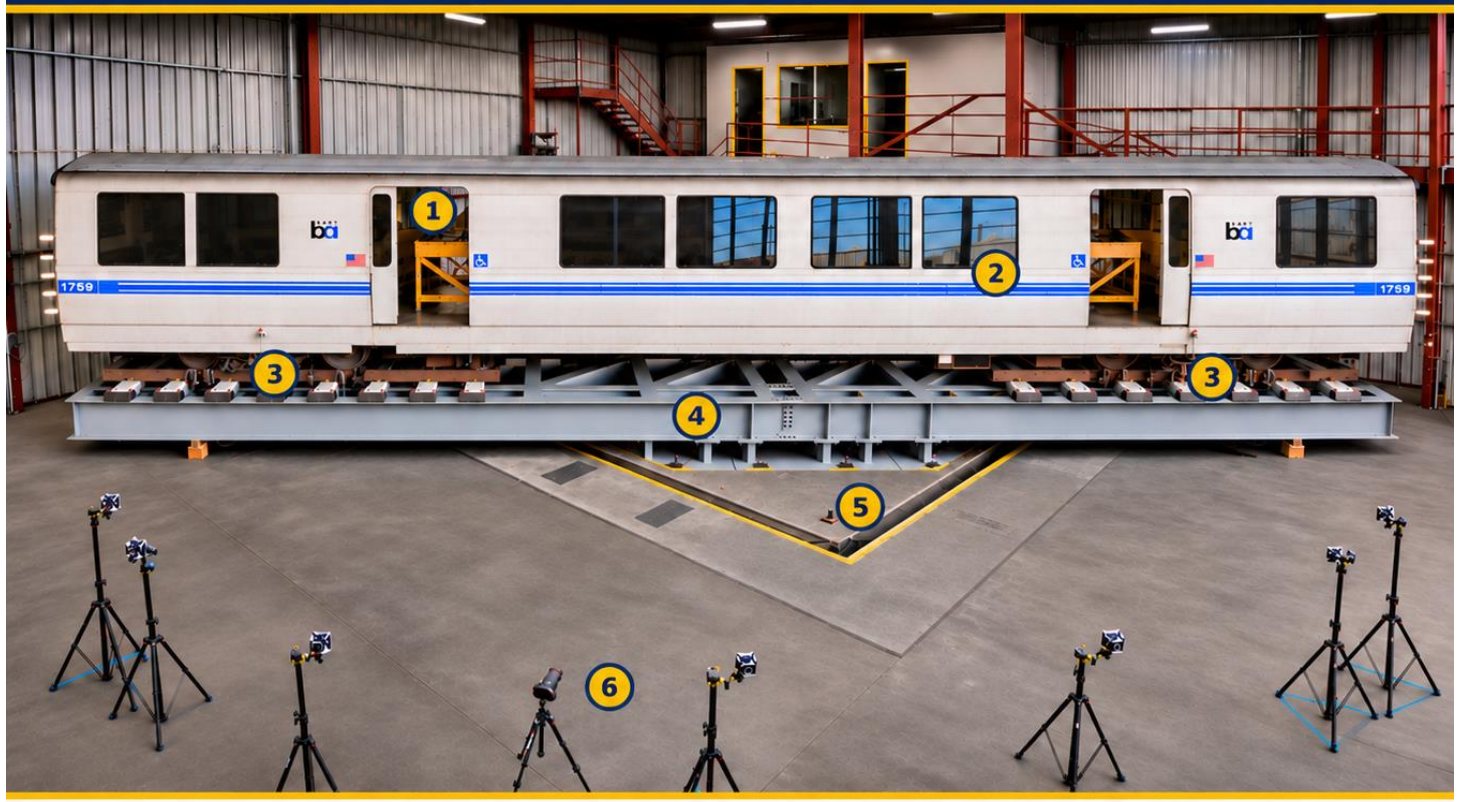
Time	Activity
11:00–11:40	Welcome, 15 min overview, setup walk through, posters and Q&A
11:40–12:10	**Live shaking-table demonstration: Set 6 + white noise**
12:10–12:30	Future collaboration discussion
12:30–1:00	Laboratory tour: Building 484 and other test specimens

Demonstration timing

Four earthquake motions ≈ 9 min; white noise ≈ 6 min; transitions ≈ 10 min.

BART Railcar | PEER Shaking-Table Testing

Full-scale BART railcar • PEER shaking table test



1 Simulated passenger-load structure
Wooden frame with distributed weights to simulate passenger loading (AW3 configuration).

2 Full-scale legacy BART railcar
Vehicle shown on the full-scale test platform (BART legacy series, Car No. 1759).

3 Rail and purpose-built instrumentation and support fixtures
Rail and purpose-built instrumentation and support fixtures below the wheelsets (short track segments).

4 Shaking-table extension platform
Structural steel frame that supports the track and railcar on the PEER shaking table.

5 PEER shaking table
Large six-degree-of-freedom (6-DOF) shaking table at the Pacific Earthquake Engineering Research (PEER) Center.

6 Optical motion-capture and video cameras
OptiTrack cameras for three-dimensional displacement tracking, complemented by a conventional video camera for test documentation.

Note: Additional instrumentation includes conventional accelerometers, PEER-CENTS inertial sensors, strain gauges, wirepot displacement sensors, and Novotechnik deformation sensors, not individually identified.

Concept and technical content: Khalid M. Mosalam | Graphic design assisted by OpenAI ChatGPT | Based on an original photograph of the BART railcar test at the PEER Center, UC Berkeley.

Why Test a BART Railcar?

- Earthquake early warning can trigger train slowdowns or stops.
- Large-scale testing helps evaluate railcar response and establish defensible shaking thresholds.
- Research balances passenger safety, infrastructure resilience, and transit operations.

The Overall PEER-BART Research Program

Literature review + numerical simulations + large-scale experiments

Evidence to assess & improve BART shaking thresholds while balancing safety and service continuity

Three complementary experimental programs support the development of evidence-based earthquake early warning thresholds.

BALLASTED TRACK

Track response and potential ballast disturbance

AERIAL STRUCTURES

Seismic damage and implications for train passage

FULL-SCALE RAILCAR

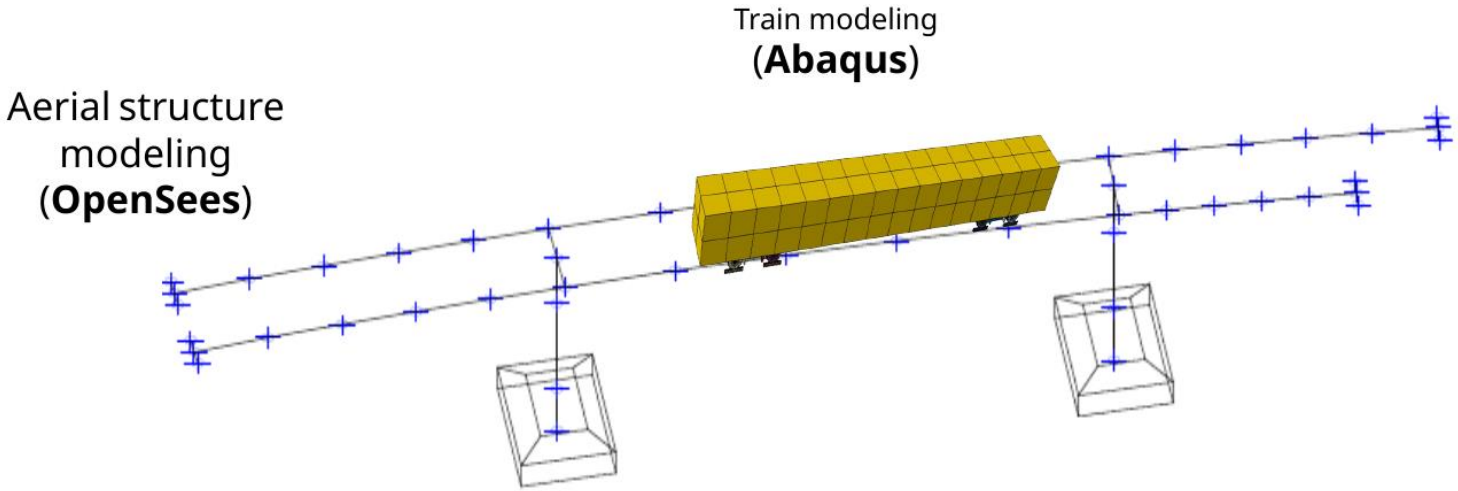
Vehicle response, rocking and potential uplift

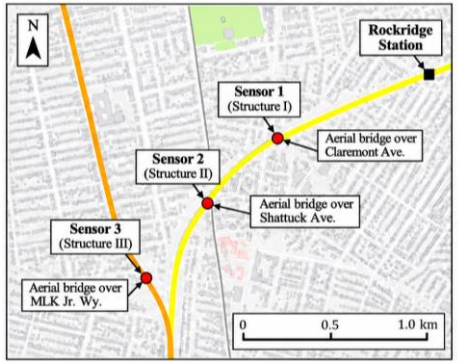
Today's live railcar demonstration is one component of this larger program.

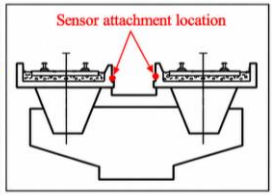
From Numerical Models to Large-Scale Experiments

Analytical and field studies

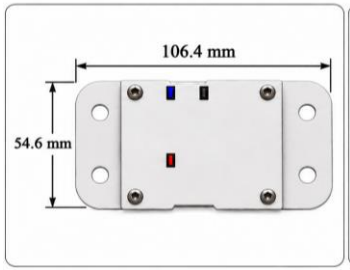
Aerial structures modeled in OpenSees and railcars in Abaqus, supported by PEER-CENTS field measurements.

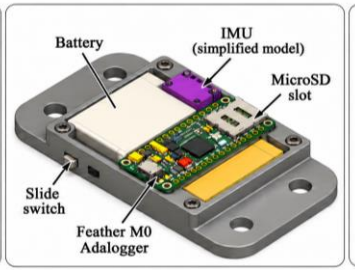


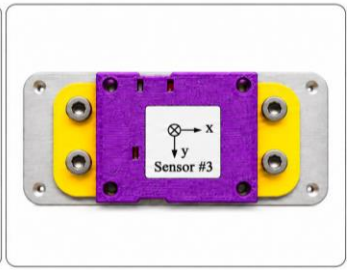












PHASE II · MODELED AERIAL SITES

Rockridge area · three structures

I · Claremont Ave. (straight)

II · Shattuck Ave. (curved)

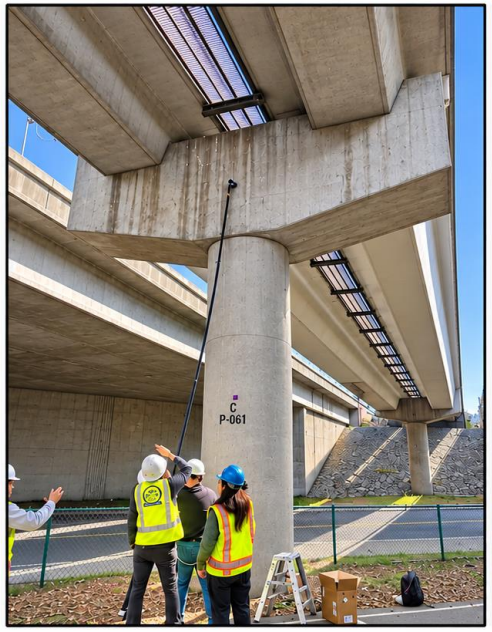
III · MLK Jr. Way (long-span)

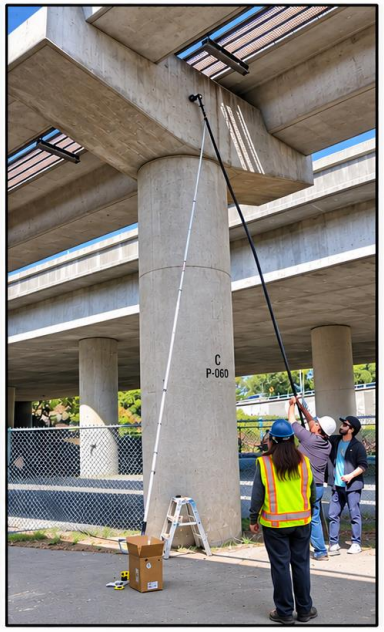
Simulations of varying earthquake intensities helped inform the experimental program, including selection of the prototype aerial structure.

BART Field Tests · PEER-CENTS and Robotic Deployment

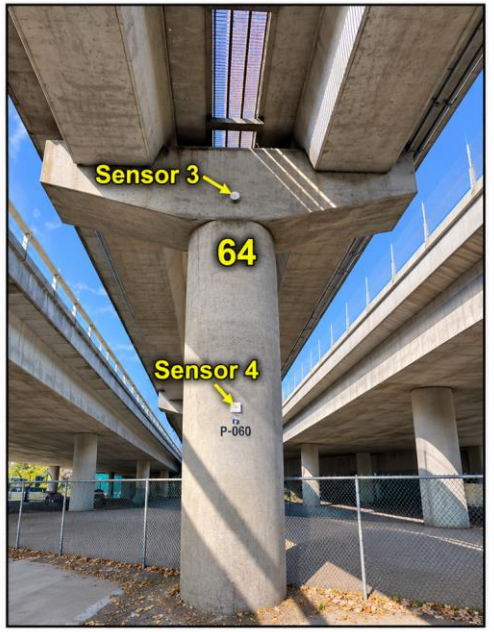
Prototype aerial structure · sensor placement and vibration measurements





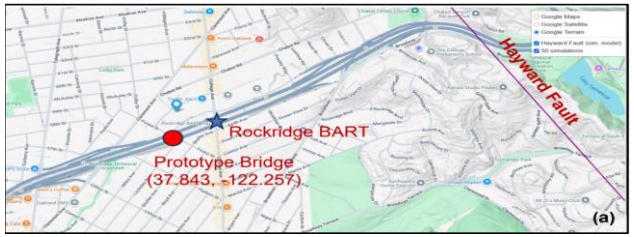






- In-house robotic arm and flexible pole used to install and retrieve PEER-CENTS units.
- Sensors placed at difficult-to-access column and cap-beam locations.
- Field vibration measurements supported identification of structural dynamic characteristics.

PROTOTYPE SITE · ROCKRIDGE



37.843° N, 122.257° W · Column 65

Earlier PEER Shaking-table Experiments

Three complementary experimental programs underpin the current railcar tests.

BALLASTED TRACK TESTS



Shaking of instrumented track and ballast to assess disturbance and track performance.

AERIAL STRUCTURE / COLUMN TESTS



Large-scale structural-component tests (***based on field tests***) to study damage, structural response, and train passage.

AW3 Loading and Scale of the Experiment

AW3 test configuration | force (kip / kN) and equivalent mass (metric tonnes)

Legacy BART railcar	60 kips	267 kN	27.2 t
Simulated seated-passenger load	7 kips	31 kN	3.2 t
Simulated standing-passenger load	30 kips	133 kN	13.6 t
Railcar + passenger loading	**97 kips**	**431 kN**	**44.0 t**
Rails and ties	11 kips	49 kN	5.0 t
Extension platform	40 kips	178 kN	18.1 t

Total test assembly: ~148 kips | ~658 kN | ~67.1 metric tonnes

PEER: Largest U.S. indoor six-degree-of-freedom shaking table; 100,000-lb specimen at 1.5g.

Berkeley Engineering (2022): Specimens up to **150,000 lb (68.0 metric tonnes)** can be tested.

Instrumentation and 3D Optical Motion Capture

Optical capture — **360 Hz** — OptiTrack PrimeX 22 cameras record reflective markers.

Marker placement — **12 markers**

4	4	4
Car body	Trucks	Support frame

Other sensors — Wire potentiometers, accelerometers, strain gauges, PEER-CENTS inertial sensors, and displacement sensors.

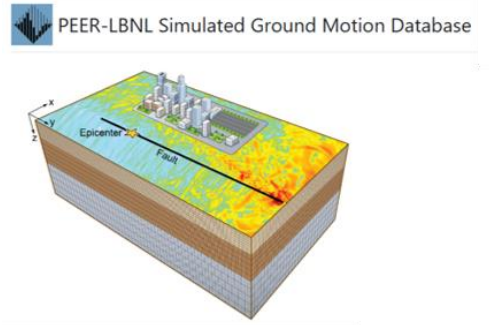
Test Matrix | Broader Program and Selected Demo and its Sequence

Previously completed in AW3: Ballasted Track Sets 1–8 (MMI 3.7–9.1) and Aerial Structure Sets 1–2 (MMI 4.2–6.0), with no damage or major issues reported.

Selected for this demo: Ballasted Track Set 6 (Level 4), four recorded/simulated rail-acceleration motions.

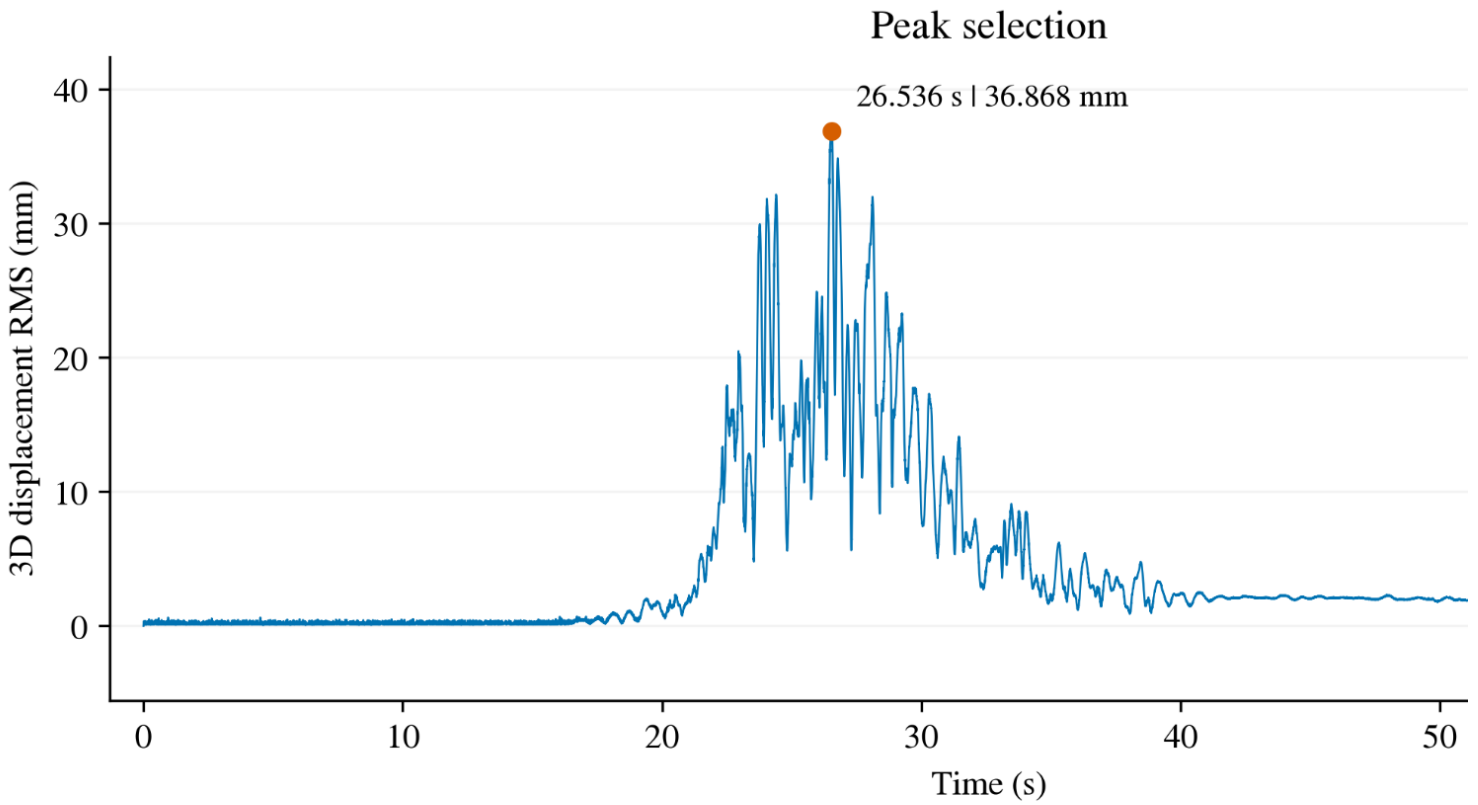
MMI range for Set 6: 6.4–7.9.

1. 1989 Loma Prieta — Los Gatos Station
2. Simulated M7.2 Hayward/M7.2 Calaveras mainshock sequence — Fremont BART Station
3. 1979 Imperial Valley — El Centro Array #8
4. **Simulated M7.0 Hayward rupture — Rockridge area (PEER–LBNL SGMD)**
5. White-noise characterization test following Set 6



OptiTrack | Preliminary Displacement Measurements

- OptiTrack records 3D marker trajectories from the railcar, trucks, and support frame.
- Common peak-response time: ~26.5 s.
- Results are preliminary.



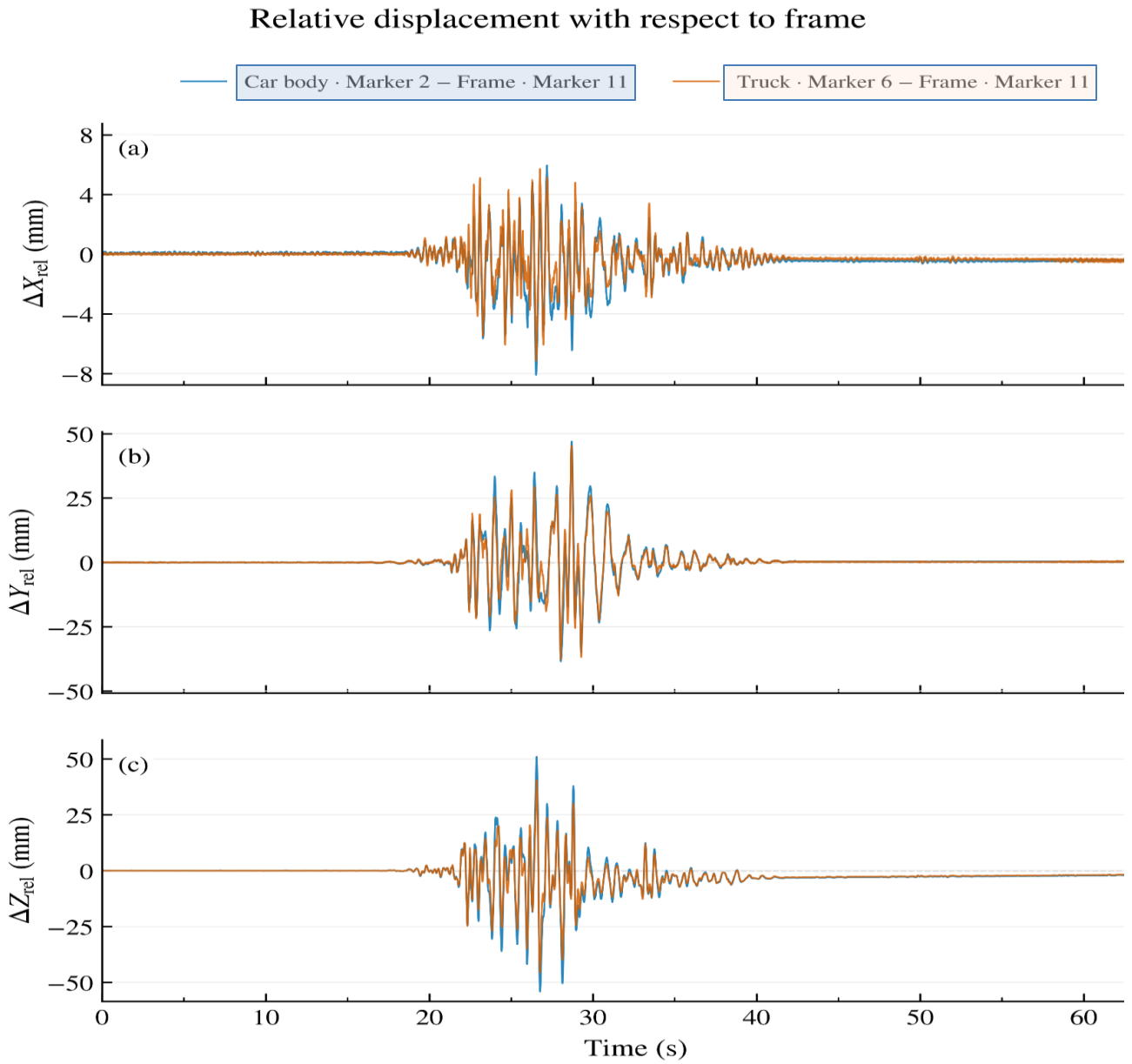
$$d_i(t) = \sqrt{[x_i(t) - x_i(t_0)]^2 + [y_i(t) - y_i(t_0)]^2 + [z_i(t) - z_i(t_0)]^2}$$

$$S(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d_i(t)^2}, \qquad N = 12, \qquad t^* = \arg\max_{t \in \mathcal{T}} S(t)$$

t_0: reference time; $\mathcal{T}$: frames with all 12 markers valid.

Relative Displacement of the Railcar and Trucks

- Relative displacements quantify how railcar & trucks move with respect to supporting frame.
- Measurements can help identify amplification & rocking.



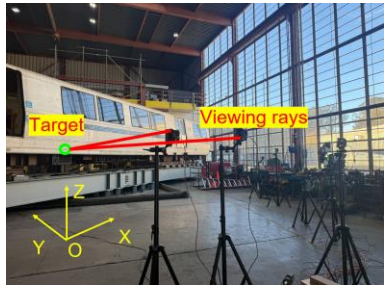
OptiTrack Demonstration Video

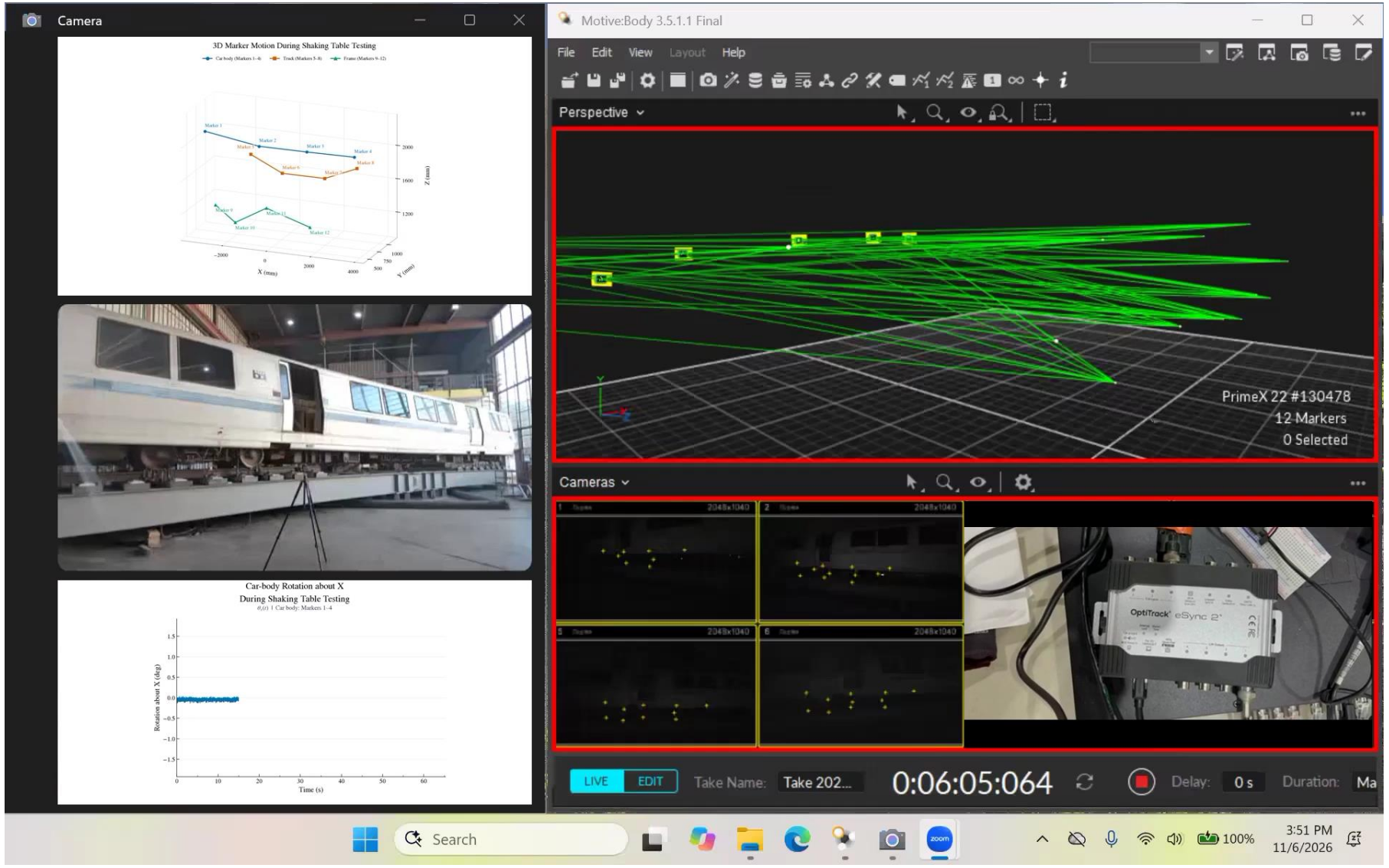
Camera viewing rays locate reflective markers in 3D.

3D marker motion during shaking:

- **Blue:** Car-body markers 1–4
- **Orange:** Truck markers 5–8
- **Green:** Support-frame markers 9–12
- **X, Y, Z (mm):** 3D marker coordinates showing relative positions & motions of measured components.

Car-body rotation about X, θ_x (degrees)





What the Test Results Will Help Us Understand

Research questions

01 Railcar motions, rocking, and potential uplift

02 Response relative to support and track

03 Sensitivity to different earthquake motions

Application for BART

Evidence supporting recommended BART earthquake operating thresholds

After the Demonstration | A Proposed PEER–BART Program

PEER-Transit — concept for discussion

A sustained research partnership could help translate seismic research into practical operating decisions for BART.

A PROVEN FRAMEWORK	Adapt the PEER–Caltrans PEER-Bridge model: A master agreement, jointly defined priorities, and competitive task orders.
EXPERTISE AND FACILITIES	Leverage PEER's earthquake engineering community, OpenSees, ground-motion resources, instrumentation, and testing facilities.
FLEXIBLE IMPLEMENTATION	Begin with a focused pilot and scale the program based on BART priorities, resources, and outcomes.

Discussion question: Which challenges would create the greatest near-term value for BART?

After the Demonstration | PEER-Transit Opportunities

Six initial topics of PEER-Transit concept

01 Aerial structures

Seismic assessment and fragility functions

02 Network resilience

Regional scenario-earthquake assessment using SGMD

03 Smart instrumentation

Sensors on aerial structures and railcars for decisions

04 Post-earthquake assessment

Field reconnaissance and AI-supported evidence

05 New railcars

Fragility functions for next-generation vehicles

06 Functional recovery

Design and operational strategies to restore service

Ideas to be refined jointly with BART; program scale and scope remain open.

Acknowledgments

BART — Project Partners

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BART

STAIRlab
STructural Artificial Intelligence Rese

Explore the Interactive BART Digital Postcard

Scan for postcard, project overview & slides.

